

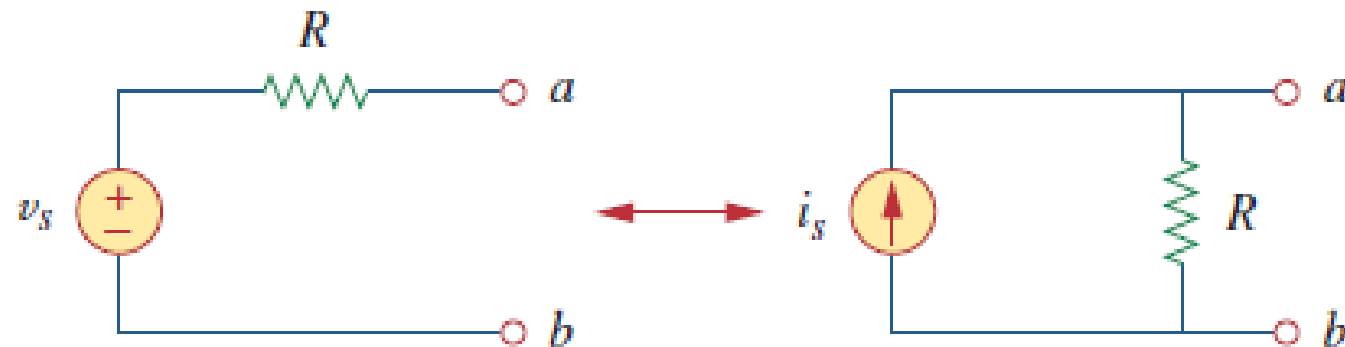
Module -1

DC Circuit Analysis and Theorems

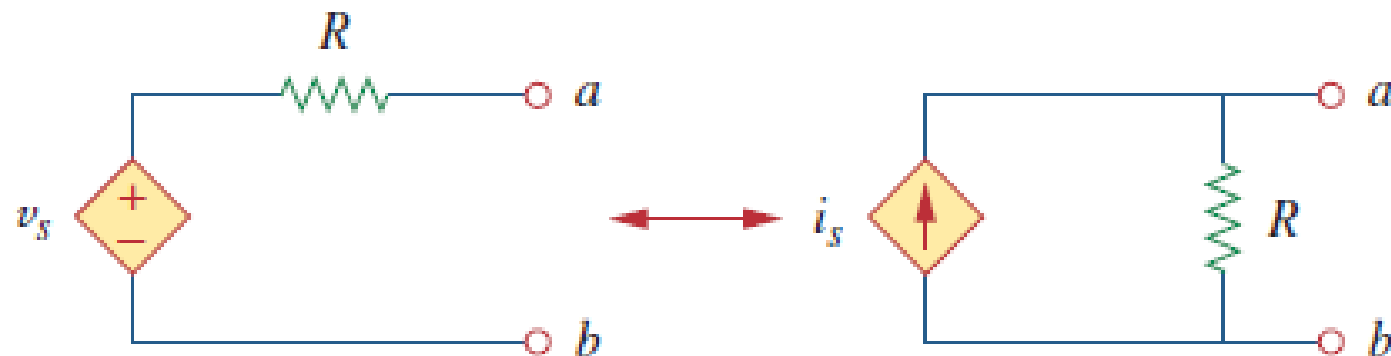
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SENSE

Source Transformation Theorem

A **source transformation** is the process of replacing a voltage source v_s in series with a resistor R by a current source i_s in parallel with a resistor R , or vice versa.



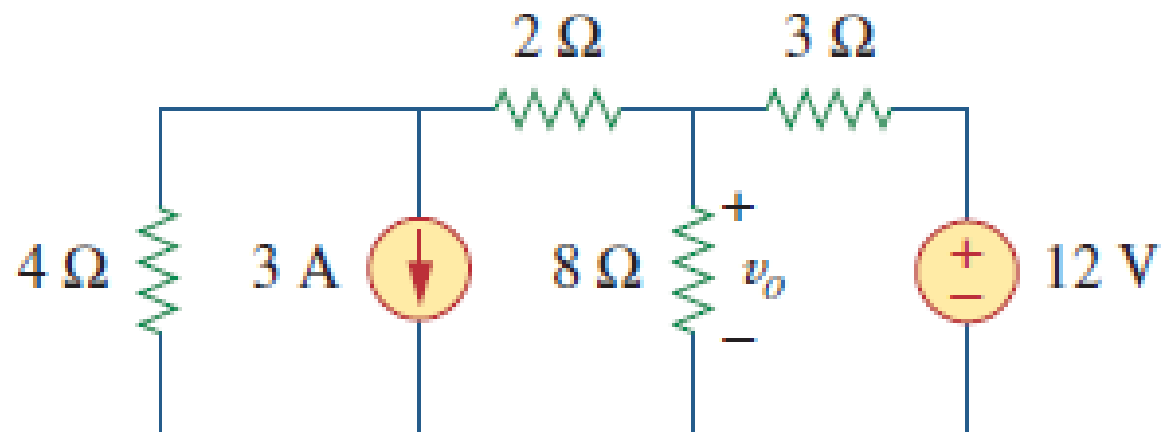
$$v_s = i_s R$$



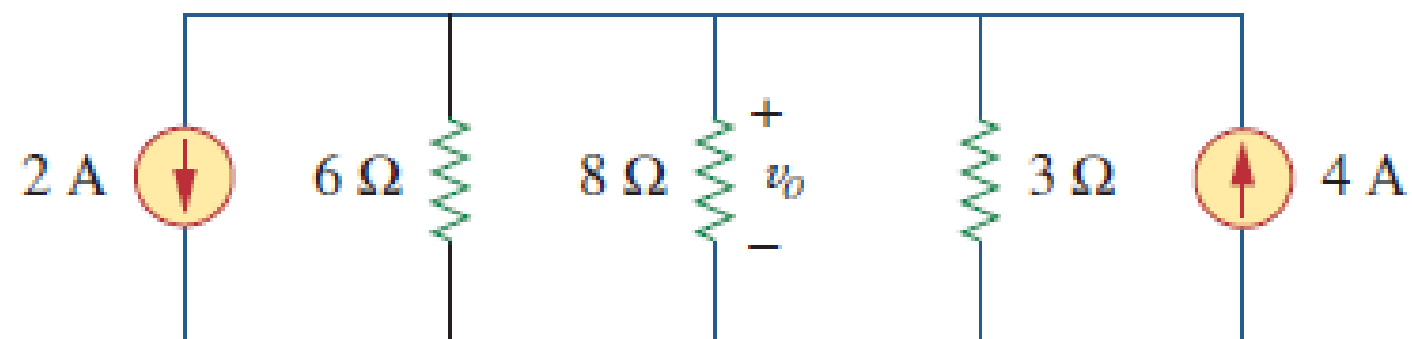
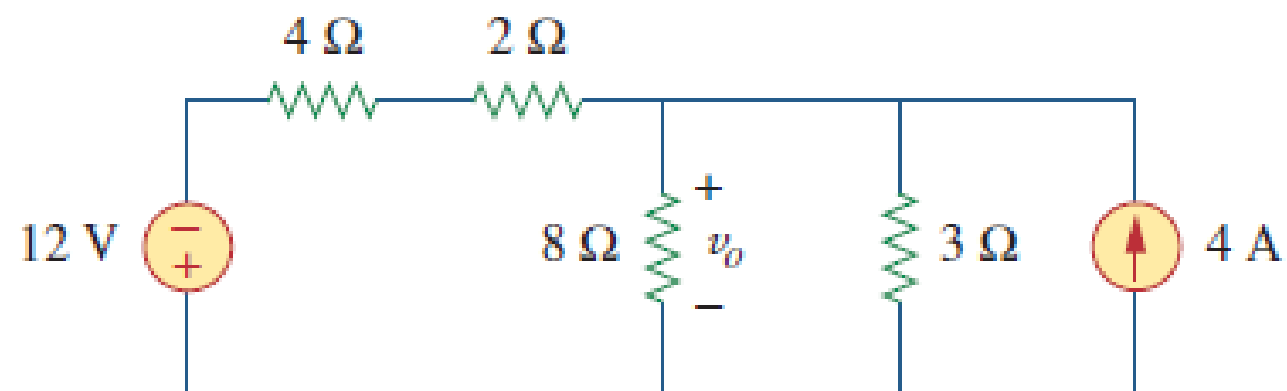
$$i_s = \frac{v_s}{R}$$

Tutorial - 1

Use the source transformation to find v_o in the circuit.

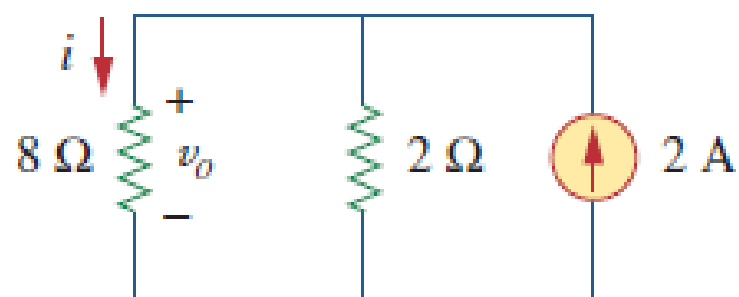


Solution:



$$i = \frac{2}{2 + 8}(2) = 0.4 \text{ A}$$

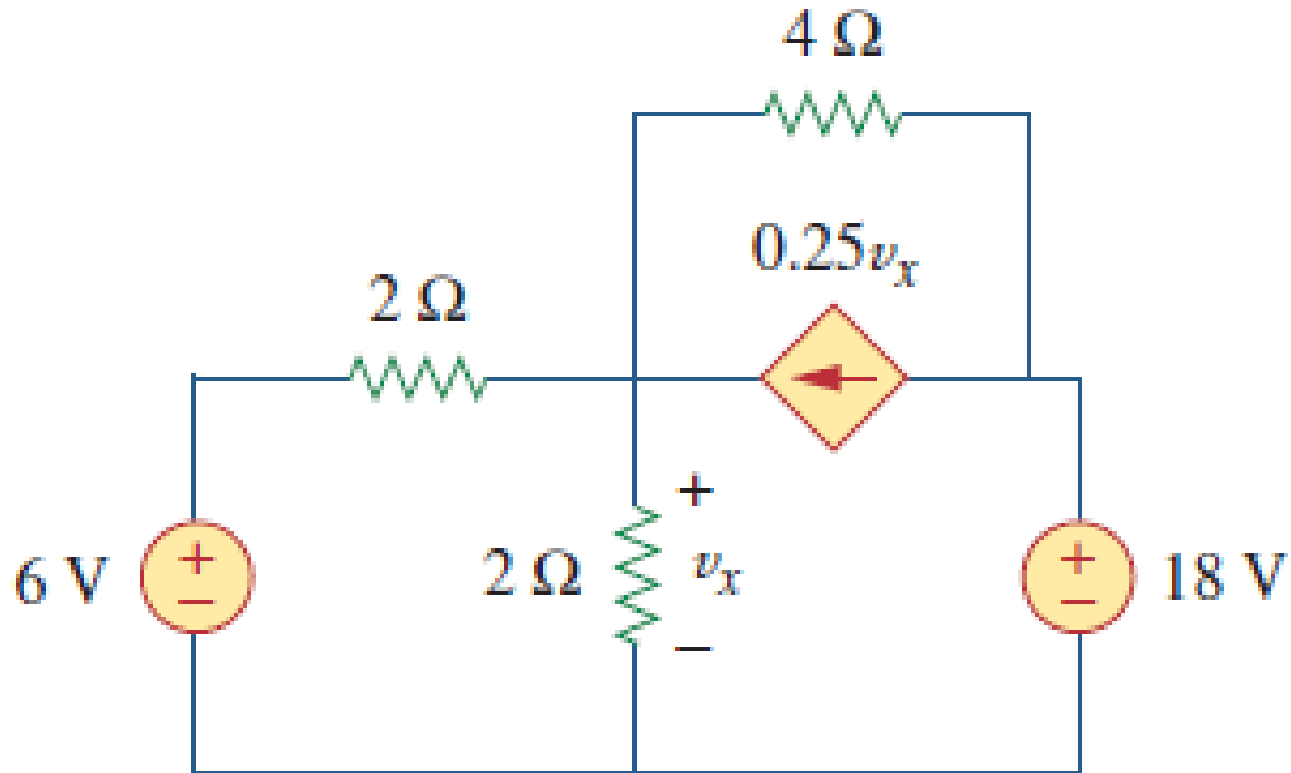
$$v_o = 8i = 8(0.4) = 3.2 \text{ V}$$



$$v_o = (8 \parallel 2)(2 \text{ A}) = \frac{8 \times 2}{10}(2) = 3.2 \text{ V}$$

Tutorial - 2

Determine v_x in the circuit using source transformation theorem.



Solution:

$$-3 + 5i + v_x + 18 = 0$$

$$-3 + 1i + v_x = 0$$

$$v_x = 3 - i$$

$$15 + 5i + 3 - i = 0$$

$$i = -4.5 \text{ A}$$

$$v_x = 3 - i = 7.5 \text{ V.}$$

